12 – H0 Test and Errors

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Recap: Type 1 and 2 Errors

Size of a Test

- If the null hypothesis H_0 is true, but **rejected**, then a **Type I Error** occurs.
- The probability of a Type I Error is

$$\mathbb{P}(H_0 \text{ rejected} | H_0).$$

- $\mathbb{P}(H_0 \text{ rejected} | H_0)$ is also called the **size** of the test.

Power of a Test

- The probability of a Type II Error is denoted by β:

$$\beta = \mathbb{P}(H_0 \text{ not rejected} | H_1)$$

- The probability that H_0 is rejected if it is wrong is the **power** of the test, i.e.

$$\textbf{Power} = \mathbb{P}(H_0 \text{ rejected} | H_1) = 1 - \beta.$$

Problem 1 Let $X_1, \ldots, X_{10} \sim N(\mu, \sigma^2)$ be i.i.d, where μ and σ are both unknown. Consider a test for $H_0 : \mu = 10$ against $H_1 : \mu \neq 10$ based on the test statistic $T = \frac{\overline{X}-\mu}{s/\sqrt{n}}$, where s^2 is the sample variance. Suppose we reject H_0 if and only if $|T| \geq t_0$.

(a) Find t_0 so that the size of the test is 0.05.

(b) Using the t_0 from part (a), is H_0 rejected for the following observations?

$$23.3,\ 3.5, -1.0,\ 40.3,\ 34.5,\ 9.6,\ 23.4,\ 18.5,\ 0.7,\ 9.0.$$

Case	Statistic Q	Dist. of Q	CI
Known σ^2	$\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$	$N(0,1)$	$\left[\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$
Unknown σ^2 ($n \geq 30$)	$\frac{\overline{X}-\mu}{S/\sqrt{n}}$	$\approx N(0,1)$	$\left[\overline{X} - z_{\alpha/2}\frac{S}{\sqrt{n}}, \overline{X} + z_{\alpha/2}\frac{S}{\sqrt{n}}\right]$
Unknown σ^2 ($n < 30$)	$\frac{\overline{X}-\mu}{S/\sqrt{n}}$	$t-$distribution degree freedom $n-1$	$\left[\overline{X} - t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}, \overline{X} + t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}\right]$

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$$t_0 = t_{0.025}(n-1) = t_{0.025}(9) = 2.262.$$

$$\overline{x} = \frac{1}{10}\sum_{i=1}^{10} x_i = 16.18, \quad s = \sqrt{\frac{1}{9}\sum_{i=1}^{10}(x_i - \overline{x})^2} \approx 14.2.$$

Thus, $t = \frac{16.18-10}{14.2/\sqrt{10}} = 1.376.$

Problem 2 Let $X_1, \ldots, X_{10}$ be an i.i.d sample drawn from $\mathrm{Exp}(\theta)$ where $\theta \in (0, \infty)$ is an unknown parameter. Consider a test for $H_0 : \theta = 1$ against $H_1 : \theta = \frac{1}{2}$ based on the test statistic $T = \sum_{i=1}^{n} X_i$.

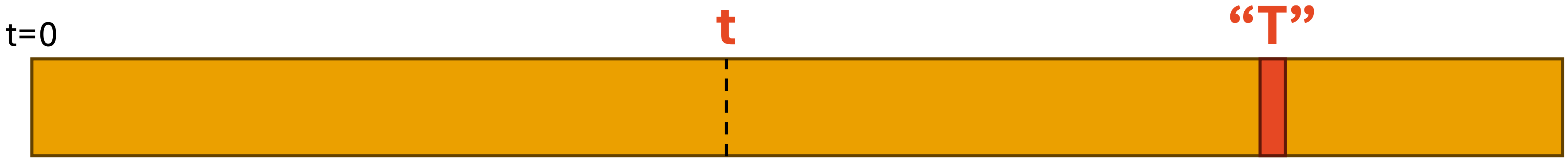
(a) Find the observed value t_0 of T such that the p-value $\mathbb{P}(T \leq t_0 | H_0)$ is equal to 0.05.

(b) Using the t_0 from part (a), consider that test that rejects H_0 if and only if $T \leq t_0$. What is the size and the power of the test?

(c) Using the test from part (b), is H_0 rejected for the following observations?

$$0.1, 0.2, 0.1, 0.3, 0.5, 0.01, 1.2, 0.05, 0.001, 0.1$$

Exponential, Poisson and Gamma – Are they related?



Consider a degenerate case where we are waiting for **1 event** to occur.

W ~ Gamma(α= 1, θ)

$$f(w) = \frac{1}{\Gamma(\alpha)\theta^{\alpha}} w^{\alpha-1} e^{-w/\theta}$$

= 1/θ exp(-w/θ)

Hence, W ~ exp(θ)

Consider a Poisson Process or Poisson Distribution with rate = λt.

X ~ Poisson(λt)

$$p(x) = e^{-\lambda} \frac{\lambda^x}{x!},$$

Suppose **T** is the waiting time until the first event occurs.
If the first event hasn't happened by time t, then clearly zero events occurred in the interval [0,t].

Pr(T > t) = Pr(X = 0) = exp(- λt)
Pr(T ≤ t) = 1 - Pr(T > t) = 1 – exp(- λt)

Which is the CDF of **exp(**θ = 1/λ**).**
Hence, **T ~ exp(**θ = 1/λ**).**

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$$0.1, 0.2, 0.1, 0.3, 0.5, 0.01, 1.2, 0.05, 0.001, 0.1$$

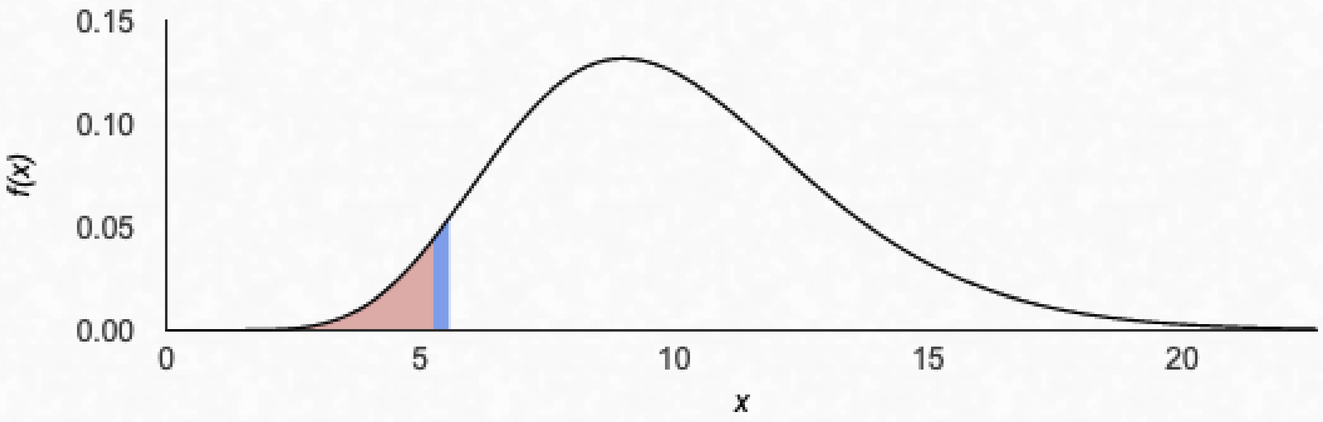
Hint: MGF of $X \sim \Gamma(\alpha, \theta)$ is $M_X(t) = (1 - \theta t)^{\alpha}$.

Gamma Distribution

$X \sim Gamma(\alpha, \beta)$

$\alpha =$ 10 Scale $\beta =$ 1 H_0

$x =$ 5.42541 $P(X < x) =$ 0.05



$\mu = E(X) = 10 \quad \sigma = SD(X) = 3.1623 \quad \sigma^2 = Var(X) = 10$

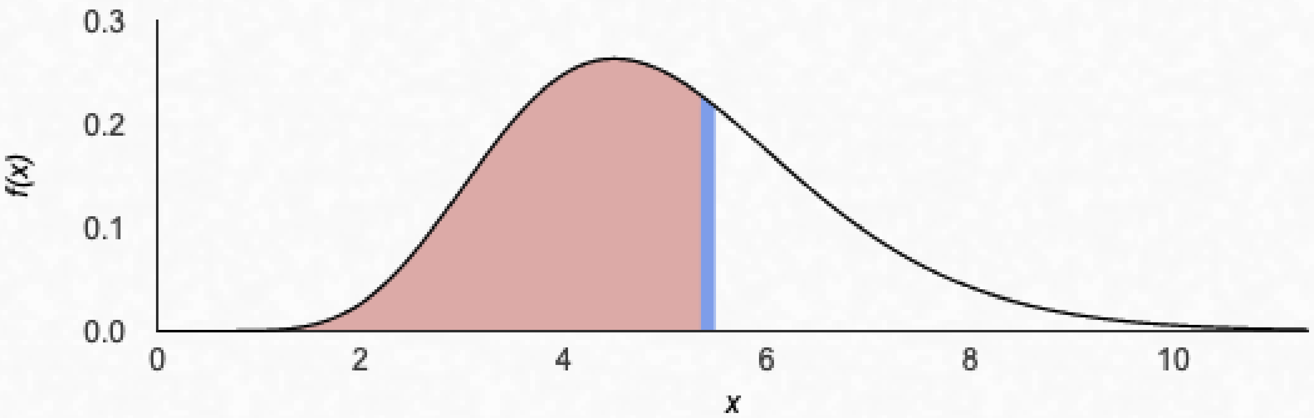
Help

Gamma Distribution

$X \sim Gamma(\alpha, \beta)$

$\alpha =$ 10 Scale $\beta =$ 1/2 H_1

$x =$ 5.424 $P(X < x) =$ 0.64277



$\mu = E(X) = 5 \quad \sigma = SD(X) = 1.5811 \quad \sigma^2 = Var(X) = 2.5$

Help

$$t = \sum_{i=1}^{10} x_i = 2.561 < 5.425.$$

Problem 3 Let $X_1, \ldots, X_5$ be an i.i.d sample drawn from Bernoulli(p) where $p \in [0,1]$ is an unknown parameter. Consider the test for $H_0 : p = 0.2$ against $H_1 : p = 0.5$ which rejects H_0 if and only if $\sum_{i=1}^{5} X_i > 2$.

(a) Compute the probabilities for Type-I and Type-II Errors

(b) Find the size and the power of the test.

Q3(a) 0.05792, 0.5 **Q3(b)** 0.05792, 0.5

Finals

- Midterms: Tut 1 to 4
- Prof have not tested **Tut 5 to 12…**

Chapters	Additional Exercises
Bivariate Distributions (Discrete and Continuous)	Exercise 4
Population & Central Limit Theorem	Exercise 5
MLE	Exercise 5
Confidence Interval	NA
Hypothesis Testing	NA

tinyurl.com/1820dropbox

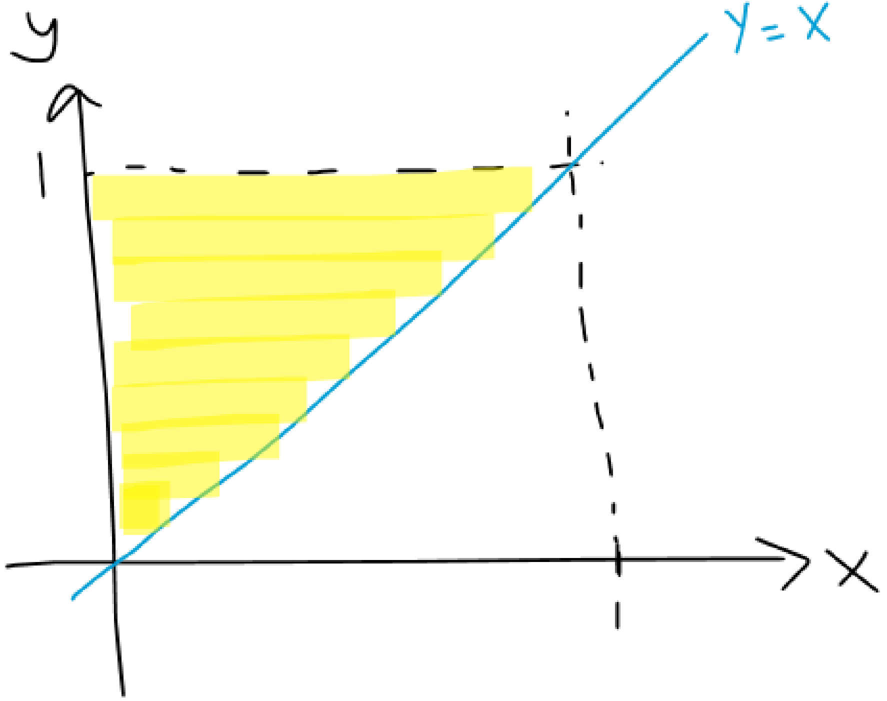
Revision: Finding the correct bounds for Continuous Integral (Tut 8)

Consider any pmf f(x,y) with $0 < x < 1$ and $0 < y < 1$.

Find $\Pr(X < Y)$

Steps:

1. Draw a graph to visualize the bounds.
2. Fill in the associated boundaries in the integral.



$$= \int_0^1 \int_x^1 f(x,y)\, dy\, dx$$

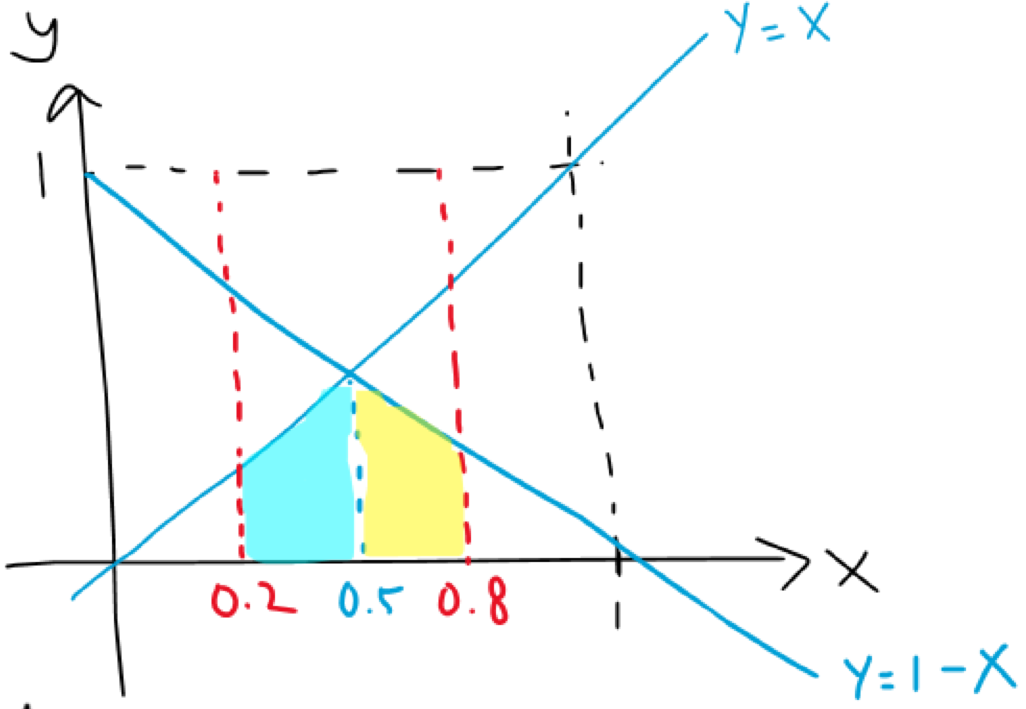
Revision: Finding the correct bounds for Continuous Integral (Tut 8)

Consider any pmf f(x,y) with 0 < x < 1 and 0 < y < 1 - x.

Find Pr(X > Y | 0.2 < X < 0.8) $= \dfrac{\Pr\{X>Y \cap 0.2<X<0.8\}}{\Pr\{0.2<X<0.8\}}$

Steps:

1. Draw a graph to visualize the bounds.
2. Fill in the associated boundaries in the integral.



$$= \int_{0.2}^{0.5}\int_{0}^{x} f(x,y)\,dy\,dx + \int_{0.5}^{0.8}\int_{0}^{1-x} f(x,y)\,dy\,dx$$